

Ueli M. Angst

Curriculum Vitae – September 2026

Personal details

First & FAMILY name:	Ueli Michael ANGST	Date of birth:	28 August 1980
Nationalities:	Swiss and French	Status:	Married, 3 children
Telephone:	+41 (0)79 944 03 40	E-mail:	uangst@ethz.ch
Researcher ID:	M-6348-2014	Google Scholar:	Link to profile
Website:	http://www.ifb.ethz.ch/durability		

Summary

Associate Professor at ETH Zurich and head of the Durability of Engineering Materials group, focusing on corrosion mechanisms, durability of infrastructure, electrochemistry, transport in porous materials, condition assessment, monitoring, and corrosion mitigation. My research combines experimental and computational methods from corrosion science, materials science, civil engineering, and reactive mass transport to improve the assessment, prediction, and service-life management of engineered materials and structures.

Key facts: 140+ scientific journal publications and 150+ conference contributions; 2 patent applications filed.

Current and previous positions

Since 06/2024	Associate professor at ETH Zurich , Switzerland, heading the research group Durability of Engineering Materials
Since 08/2021	Co-founder of ETH-SpinOff DuraMon
Since 04/2017	President of the Swiss Society for Corrosion Protection (SGK), which is the leading agency in corrosion consulting in Switzerland (and which is active worldwide)
Since 04/2012	Member of the board of directors at the material testing institute IMP Bautest AG, Switzerland
01/2017–05/2024	Assistant professor at ETH Zurich , Switzerland
08/2011 – 12/2016	Postdoctoral researcher at the Institute for Building Materials , ETH Zurich , Switzerland (part time, var. 40-60%), with Prof. Dr. R.J. Flatt and Prof. Dr. B. Elsener. <i>Research, teaching, and supervising in the field of durability of concrete structures.</i>
08/2011 – 11/2016	Consulting engineer at the Swiss Society for Corrosion Protection (SGK), Zurich, Switzerland (part time, 40%). <i>Consultant for corrosion in the construction industry and energy sector; various research projects.</i>

Education

04/2007 – 05/2011	PhD thesis at the Norwegian University of Science and Technology , NTNU, Trondheim, Norway (PhD defence: 03 May 2011). Supervisor: Prof. Ø. Vennesland Topic: Chloride-induced reinforcement corrosion in concrete
10/2000 – 01/2005	Studies in civil engineering at ETH Zurich , Switzerland (diploma equivalent to MSc degree)

Awards and recognitions (selection, for full list see [research group website](#))

ERC Starting Grant (2019); **RILEM Robert L'Hermite Medal** (2017); **SNSF Professorship Grant** (2016); 10+ Best/Outstanding paper awards (Guy Bengough, Le Chatelier, etc.); best doctoral dissertation award of the German Chemical Society GDCh (2012); **Dandelion Entrepreneurship Award (2021)** by ETH Entrepreneur Club & AI Center recognizing professors for their efforts to promote entrepreneurship.

Editorial work

Editor: *CEMENT* (Elsevier, since 2020); **Associate editor:** *RILEM Technical Letters* (since 2017) and *Materials & Structures* (since 2026); **Editorial board member:** *Corrosion Communications* (since 2023); Guest editor of special issues in *Transport in Porous Media*, *CEMENT*, and *Materials and Corrosion*.

Scientific society memberships and committee work

- [RILEM](#) member and deputy chair of TC [DLC](#) (since 2026), deputy chair of TC [296-ECS](#) (2020–2026) chair of TC [262-SCI](#) (2014–2021). Member of various RILEM TCs.
- Chair of the GfKORR [working group](#) “Korrosionsschutz im konstruktiven Ingenieurbau” (since 2019).
- Steering committee of the [Swiss Corrosion Science Network](#); member of [fib](#), [DGZfP](#), [ACI](#), and [AMPP](#).

Service to the academic community and supervision

Management, faculty service, supervision

- Since 2024, **Director of Studies** for the BSc, MSc, and PhD programs in civil engineering at ETH Zurich, responsible for teaching operations, quality management, and program development.
- Supervised 14 completed PhD theses, including 11 as main advisor; supervised 80+ BSc/MSc theses and served as examiner for 25+ PhD theses internationally.

Outreach and innovation

Co-founder of ETH spin-off [DuraMon](#) and host laboratory of ETH spin-off [Talpa Inspection](#); two European patent applications filed on corrosion assessment and sensor technologies.

Publications (selection)

Researcher ID: [M-6348-2014](#); Google Scholar: [Link to profile](#); Scopus: [ID 22978555600](#)

Five most important publications (from the last 15 years)

- Albert C, Schmid T, Zhang Z, **Angst U**. Rethinking concrete durability for low-carbon concretes through climate-informed corrosion modelling. *Nature Communications* (2026) [doi.org/10.1038/s41467-026-76916-3](#)
- Governo S, Rossi E, Azad S, Kaestner A, **Angst U**. The interdependent roles of chloride, moisture and dissolved oxygen in macro-voids at the steel-concrete interface in corrosion. *Corrosion Science* 265 (2026) 113814. [doi.org/10.1016/j.corsci.2026.113814](#)
- Albert C, Mundra S, Ferreira Sanchez D, Furcas F, Rajyaguru AD, Isgor B, Grolimund D, **Angst U**. Microscale chemical imaging to characterize and quantify corrosion processes at the metal-electrolyte interface. *npj Materials Degradation* 8 (2024) 116. [doi.org/10.1038/s41529-024-00534-x](#)
- Stefanoni M, **Angst U**, Elsener B. Kinetics of electrochemical dissolution of metals in porous media. *Nature Materials* (2019). [doi:10.1038/s41563-019-0439-8](#)
- **Angst U**, Elsener B. The size effect in corrosion greatly influences the predicted life span of concrete infrastructures. *Science Advances* 3 (2017) e170075. [doi.org/10.1126/sciadv.1700751](#)

Five recent for the project relevant publications (from the last 5 years)

- Furcas F, Vogel F, Lothenbach B, Angst U. Mechanism of crevice corrosion in historic riveted steel structures. *Communications Engineering* (2026). [doi.org/10.1038/s44172-026-00780-8](#)
- Yilmaz D, Thoma K, **Angst U**. A probabilistic framework for modelling chloride-induced corrosion and related load-deformation behaviour of reinforced concrete. *Structural Concrete* (2026). [doi.org/10.1002/suco.70776](#)
- **Angst U**, Rossi E, Boschmann Käthler C, Mannes D, Trtik P, Elsener B, Zhou Z, Strobl M. Chloride-induced corrosion of steel in concrete – insights from bimodal neutron and X-ray microtomography combined with ex-situ microscopy. *Materials and Structures* 57 (2024) 56. [doi.org/10.1617/s11527-024-02337-7](#)
- Pfändler P, Bodie K, Crotta G, Pantic M, Siegwart R, **Angst U**. Non-destructive corrosion inspection of reinforced concrete structures using an autonomous flying robot. *Automation in Construction* 158 (2024) 105241. [doi.org/10.1016/j.autcon.2023.105241](#)
- Pfändler P, Bircher L, **Angst U**. Inspecting the corrosion state of underground reinforced concrete structures. *Journal of Infrastructure Preservation and Resilience* 3 (2022) 17. [doi.org/10.1186/s43065-022-00064-3](#)